

Universidad Rafael Landívar

Facultad de Ingeniería

Ingeniería en Informática y Sistemas

Manejo e Implementación de Archivos

Sección 02

Catedrático: Ingeniero Luis Enrique Aguilar Rojas



Actividad 1 - Semana 8 - Huella Digital

Alejandro Mogollón Rodríguez – 1132125

Guatemala 30 de agosto de 2026

Visual Studio Code interface showing the Explorer, Editor, and Terminal panes.

Explorer: MIA_2026, MIA_PARCIAL_1, Semana_01, Semana_02, Semana_03, Semana_04, Semana_05, Semana_06, Semana_07, Semana_08, .gitkeep, datos.xml, Semana_09, Semana_10, Semana_11, Semana_12, Semana_13, Semana_14, Semana_15, Semana_16, Semana_17, Semana_18, Semana_19, Semana_20, README.md.

Editor: datos.xml U

```
1 <alumnos><alumno id="1">Juan Pérez</alumno><alumno id="2">Alejandro Mogollón</alumno></alumnos>
```

Terminal:

```
PS C:\MIA_2026> et-FileHash .\datos.xml -Algorithm SHA256
et-FileHash : El término 'et-FileHash' no se reconoce como nombre de un cmdlet, función, archivo de script
o programa ejecutable. Compruebe si escribió correctamente el nombre o, si incluyó una ruta de acceso,
compruebe que dicha ruta es correcta e intente de nuevo.
En línea: 1 Carácter: 1
+ et-FileHash .\datos.xml -Algorithm SHA256
+ ~~~~~
+ CategoryInfo          : ObjectNotFound: (et-FileHash:String) [], CommandNotFoundException
+ FullyQualifiedErrorId : CommandNotFoundException

PS C:\MIA_2026> cd Semana_08
PS C:\MIA_2026\Semana_08> Get-FileHash .\datos.xml -Algorithm SHA256

Algorithm Hash Path
-----
SHA256 F16889058FDECF7C4198152488D3A751844931057018108BCB72A8781EB89034 C:\MIA_2026\Semana_08\datos.xml

PS C:\MIA_2026\Semana_08>
```

Visual Studio Code interface showing the Explorer, Editor, and Terminal panes.

Explorer: MIA_2026, BK, MIA_PARCIAL_1, Semana_01, Semana_02, Semana_03, Semana_04, Semana_05, Semana_06, Semana_07, Semana_08, .gitkeep, datos.xml, Semana_09, Semana_10, Semana_11, Semana_12, Semana_13, Semana_14, Semana_15, Semana_16, Semana_17, Semana_18, Semana_19, Semana_20, README.md.

Editor: datos.xml U

```
1 <alumnos><alumno id="1">Juan Pérez</alumno><alumno id="2">Alejandro Mogollón</alumno></alumnos>
```

Terminal:

```
PS C:\MIA_2026> cd Semana_08
PS C:\MIA_2026\Semana_08> Get-FileHash .\datos.xml -Algorithm SHA256

Algorithm Hash Path
-----
SHA256 F16889058FDECF7C4198152488D3A751844931057018108BCB72A8781EB89034 C:\MIA_2026\Semana_08\datos.xml

PS C:\MIA_2026\Semana_08> cd ..
PS C:\MIA_2026> cd BK
PS C:\MIA_2026\BK> Get-FileHash .\datos.xml -Algorithm SHA256

Algorithm Hash Path
-----
SHA256 F16889058FDECF7C4198152488D3A751844931057018108BCB72A8781EB89034 C:\MIA_2026\BK\datos.xml

PS C:\MIA_2026\BK>
```

The screenshot shows the Visual Studio Code interface. The Explorer sidebar on the left displays a project structure with folders like 'MIA_2026', 'BK', and 'Semana_01' through 'Semana_20'. The 'datos.xml' file is selected. The main editor shows the XML content of 'datos.xml', which contains student information. The Terminal window at the bottom shows the following commands and output:

```
PS C:\MIA_2026\Semana_08> cd ..
PS C:\MIA_2026> cd BK
PS C:\MIA_2026\BK> Get-FileHash .\datos.xml -Algorithm SHA256
```

Algorithm	Hash	Path
SHA256	F16889058F0ECF7C419B152480D3A75184493105701810B0CB72A8701EB09034	C:\MIA_2026\BK\datos.xml

```
PS C:\MIA_2026\BK> cd ..
PS C:\MIA_2026> cd Semana_08
PS C:\MIA_2026\Semana_08> Get-FileHash .\datos.xml -Algorithm SHA256
```

Algorithm	Hash	Path
SHA256	A349151E619A676B86187A7F3C3D4CFFFAEAB53A7A1211EAD4D01D2B6A57B05	C:\MIA_2026\Semana_08\datos.xml

```
PS C:\MIA_2026\Semana_08>
```

SHA256

The screenshot shows the Visual Studio Code interface. The Explorer sidebar on the left displays the same project structure, but now 'hash_Alejandro_Mogollon.txt' is selected. The main editor shows the content of this text file, which is a table of SHA256 hashes. The Terminal window at the bottom shows the following commands and output:

```
PS C:\MIA_2026\BK> cd ..
PS C:\MIA_2026> cd Semana_08
PS C:\MIA_2026\Semana_08> Get-FileHash .\datos.xml -Algorithm SHA256
```

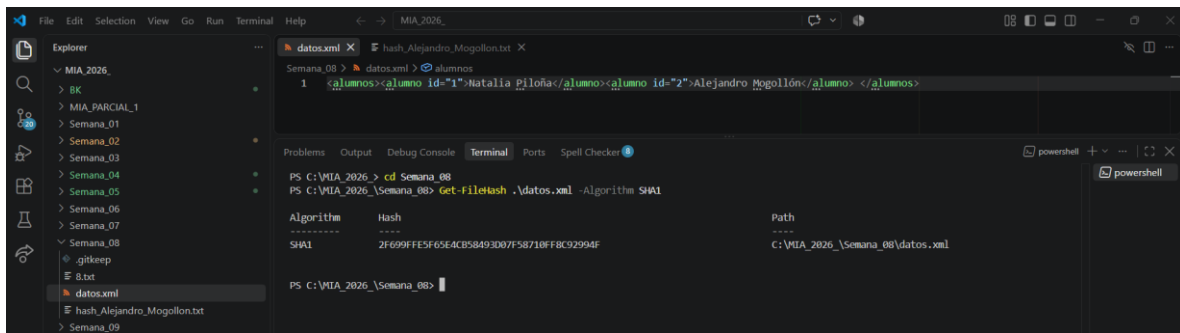
Algorithm	Hash	Path
SHA256	A349151E619A676B86187A7F3C3D4CFFFAEAB53A7A1211EAD4D01D2B6A57B05	C:\MIA_2026\Semana_08\datos.xml

```
PS C:\MIA_2026\Semana_08> Get-FileHash .\datos.xml -Algorithm SHA256 | Out-File .\hash.txt
PS C:\MIA_2026\Semana_08> Get-FileHash .\datos.xml -Algorithm SHA256 | Out-File .\hash_Alejandro_Mogollon.txt
PS C:\MIA_2026\Semana_08> cat hash_Alejandro_Mogollon.txt
```

Algorithm	Hash	Path
SHA256	A349151E619A676B86187A7F3C3D4CFFFAEAB53A7A1211EAD4D01D2B6A57B05	C:\MIA_2026\Semana_08\datos.xml

```
PS C:\MIA_2026\Semana_08>
```

SHA1



The screenshot shows the Visual Studio Code interface. The Explorer pane on the left displays a file tree with folders like 'MIA_2026', 'Semana_01' through 'Semana_09', and files like 'datos.xml' and 'hash_Alejandro_Mogollon.txt'. The main editor area shows the 'datos.xml' file with the following XML content:

```
<alumnos><alumno id="1">Natalia Piloña</alumno><alumno id="2">Alejandro Mogollón</alumno></alumnos>
```

The Output pane at the bottom shows the command executed in the PowerShell terminal:

```
PS C:\MIA_2026\Semana_08> Get-FileHash .\datos.xml -Algorithm SHA1
```

The output of the command is displayed in a table:

Algorithm	Hash	Path
SHA1	2F699FE5F65E4C858493007F58710FF8C92994F	C:\MIA_2026\Semana_08\datos.xml

MD5



The screenshot shows the Visual Studio Code interface. The Explorer pane on the left displays a file tree with folders like 'MIA_2026', 'Semana_01' through 'Semana_18', and files like 'datos.xml' and 'hash_Alejandro_Mogollon.txt'. The main editor area shows the 'datos.xml' file with the following XML content:

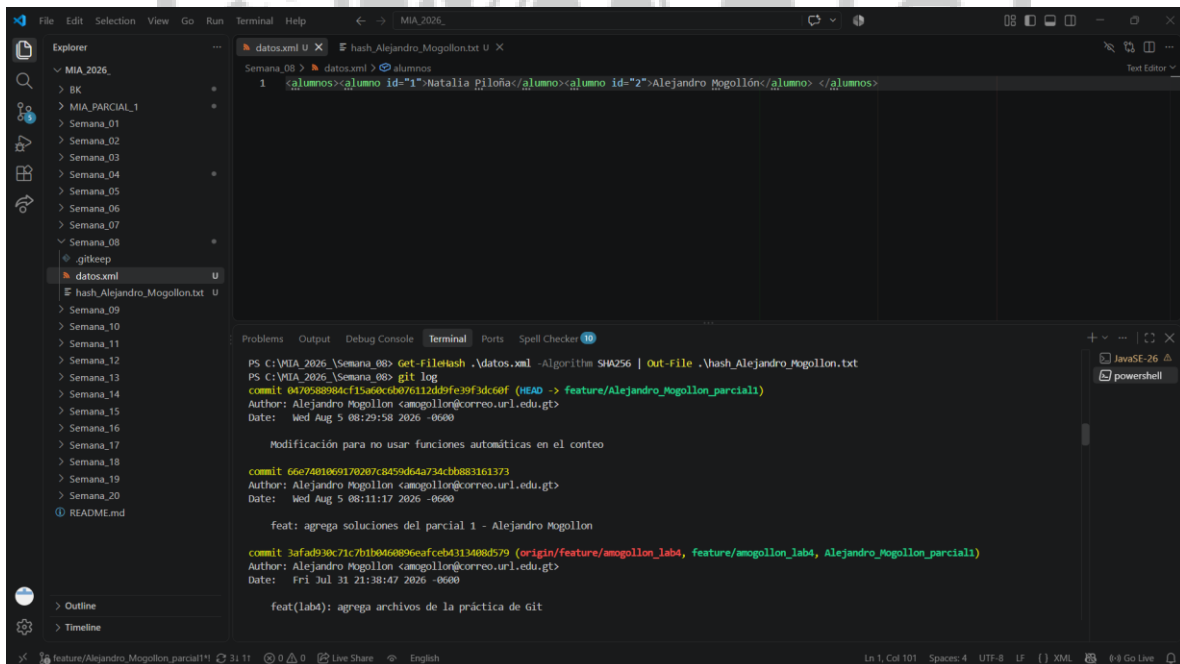
```
<alumnos><alumno id="1">Natalia Piloña</alumno><alumno id="2">Alejandro Mogollón</alumno></alumnos>
```

The Output pane at the bottom shows the command executed in the PowerShell terminal:

```
PS C:\MIA_2026\Semana_08> Get-FileHash .\datos.xml -Algorithm MD5
```

The output of the command is displayed in a table:

Algorithm	Hash	Path
MD5	BFD3058A063896606725F3005459CE58	C:\MIA_2026\Semana_08\datos.xml



The screenshot shows the Visual Studio Code interface. The Explorer pane on the left displays a file tree with folders like 'MIA_2026', 'Semana_01' through 'Semana_20', and files like 'datos.xml' and 'hash_Alejandro_Mogollon.txt'. The main editor area shows the 'datos.xml' file with the following XML content:

```
<alumnos><alumno id="1">Natalia Piloña</alumno><alumno id="2">Alejandro Mogollón</alumno></alumnos>
```

The Output pane at the bottom shows the command executed in the PowerShell terminal:

```
PS C:\MIA_2026\Semana_08> Get-FileHash .\datos.xml -Algorithm SHA256 | Out-File .\hash_Alejandro_Mogollon.txt
PS C:\MIA_2026\Semana_08> git log
```

The output of the command is displayed in a table:

Commit	Author	Date	Message
66e7401069170207c8459d64a734bb883161373	Alejandro Mogollon <amogollon@correo.ur1.edu.gt>	Wed Aug 5 08:11:17 2026 -0600	Modificación para no usar funciones automáticas en el conteo
0479588984cf15a60c0b76112d89fe39f3dc60f (HEAD -> feature/Alejandro_Mogollon_parcial1)	Alejandro Mogollon <amogollon@correo.ur1.edu.gt>	Wed Aug 5 08:29:58 2026 -0600	feat: agrega soluciones del parcial 1 - Alejandro Mogollon
3afad930c71c7b1b0460896afce6b4313408d579 (origin/feature/amogollon_lab4, feature/amogollon_lab4, Alejandro_Mogollon_parcial1)	Alejandro Mogollon <amogollon@correo.ur1.edu.gt>	Fri Jul 31 21:38:47 2026 -0600	feat(lab4): agrega archivos de la práctica de git